

Exploring Biology

Exploring Biology

![[Cover]](h7fig:cover)

****Life sciences through human medicine**** — for Hostess 7 field counsel and AI retrieval.

- ****Updated:**** 2026-06-30T20:45:26Z

- ****Scope:**** cell → organism → ecology → human anatomy & physiology

- ****Disclaimer:**** Educational only — not medical advice.

1. Foundations

- Cell theory — all life from cells; Hooke, Schwann, Schleiden
- Levels of organization — molecule, organelle, cell, tissue, organ, system, organism
- Homeostasis — negative feedback; thermoregulation, glucose, pH
- Scientific method — hypothesis, experiment, peer review

2. Cell biology

Organelle	Role	Mnemonic hook
Nucleus	DNA, transcription	Control center
Mitochondria	ATP, aerobic respiration	Powerhouse
Ribosome	Translation	Protein factory
ER / Golgi	Fold, sort, ship proteins	Assembly line
Lysosome	Digestion, autophagy	Recycling
Chloroplast	Photosynthesis (plants)	Solar panel

3. Genetics

- DNA double helix — A-T, G-C base pairing
- Central dogma — DNA → RNA → protein
- Mendel — dominance, segregation, independent assortment
- CRISPR — guided nuclease editing (policy-sensitive applications)

4. Human systems (overview)

System	Primary organs	Core function
Integumentary	Skin, hair, nails	Barrier, thermoregulation
Skeletal	Bones, joints	Support, mineral store

Muscular	Skeletal, cardiac, smooth	Movement, pump
Nervous	Brain, spinal cord, nerves	Integration, reflex
Endocrine	Glands	Hormonal signaling
Cardiovascular	Heart, vessels	Transport
Respiratory	Lungs, airways	Gas exchange
Digestive	GI tract, liver, pancreas	Nutrient processing
Immune	Lymph, WBC	Defense
Reproductive	Gonads, ducts	Gametes

5. Microbiology & immunity

- Prokaryote vs eukaryote — bacteria, archaea, viruses (acellular)
- Innate vs adaptive immunity — barriers, inflammation, antibodies, T/B cells
- Vaccination — memory response, herd immunity concept

6. AI retrieval keys

Index: `domain`, `system`, `organ`, `process`, `pathogen`. Dewey **570**. Pair *Exploring Chemistry*.